Multi-Agent Systems (MAS)

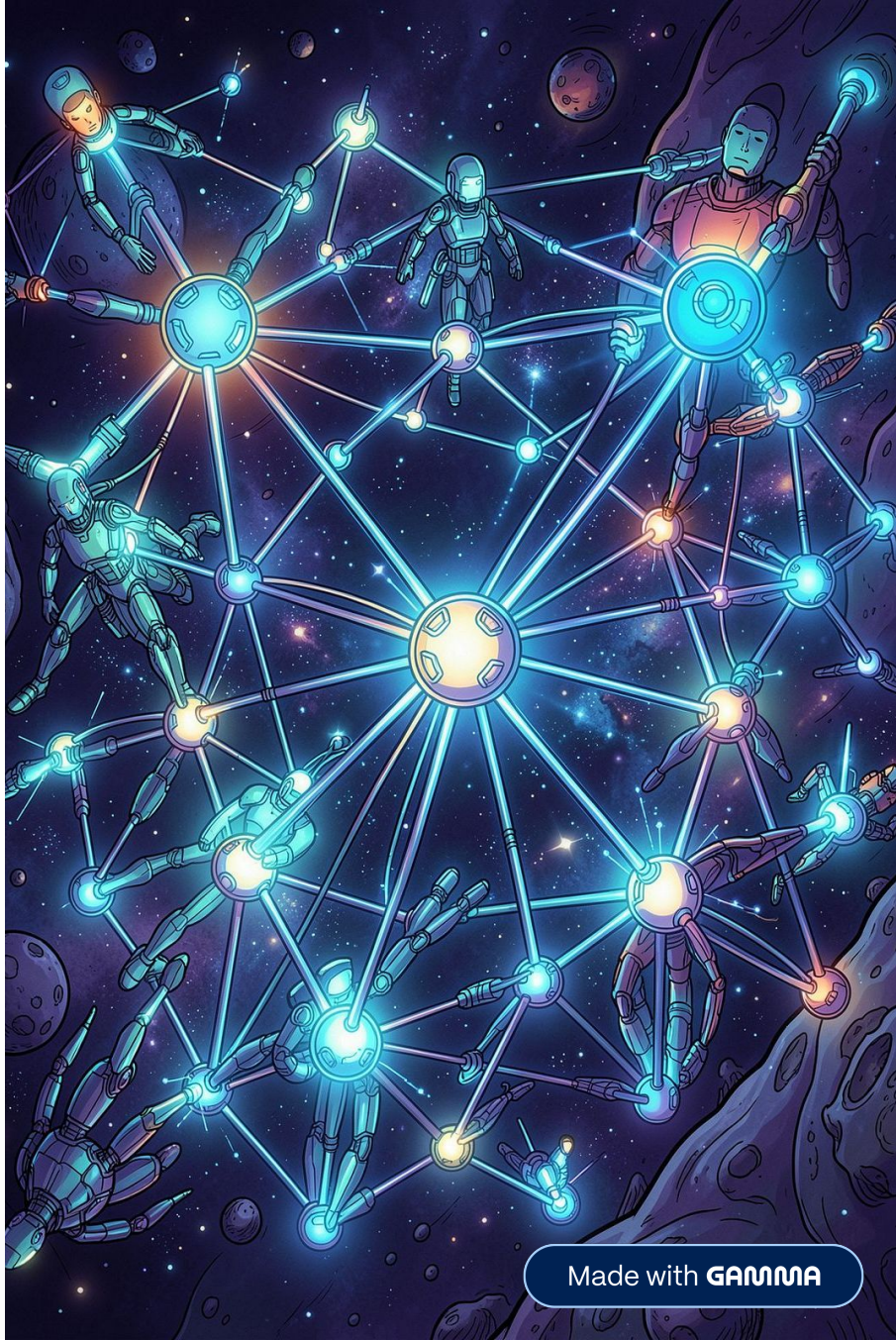
An introduction to intelligent, collaborative computing systems

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Introduction to Multi-Agent Systems

Traditional software runs as a single, centralised program. But real-world problems — from traffic management to financial trading — demand **distributed, intelligent, and collaborative** solutions. Multi-Agent Systems answer that call by modelling complex interactions between autonomous entities.

🌐 Distributed

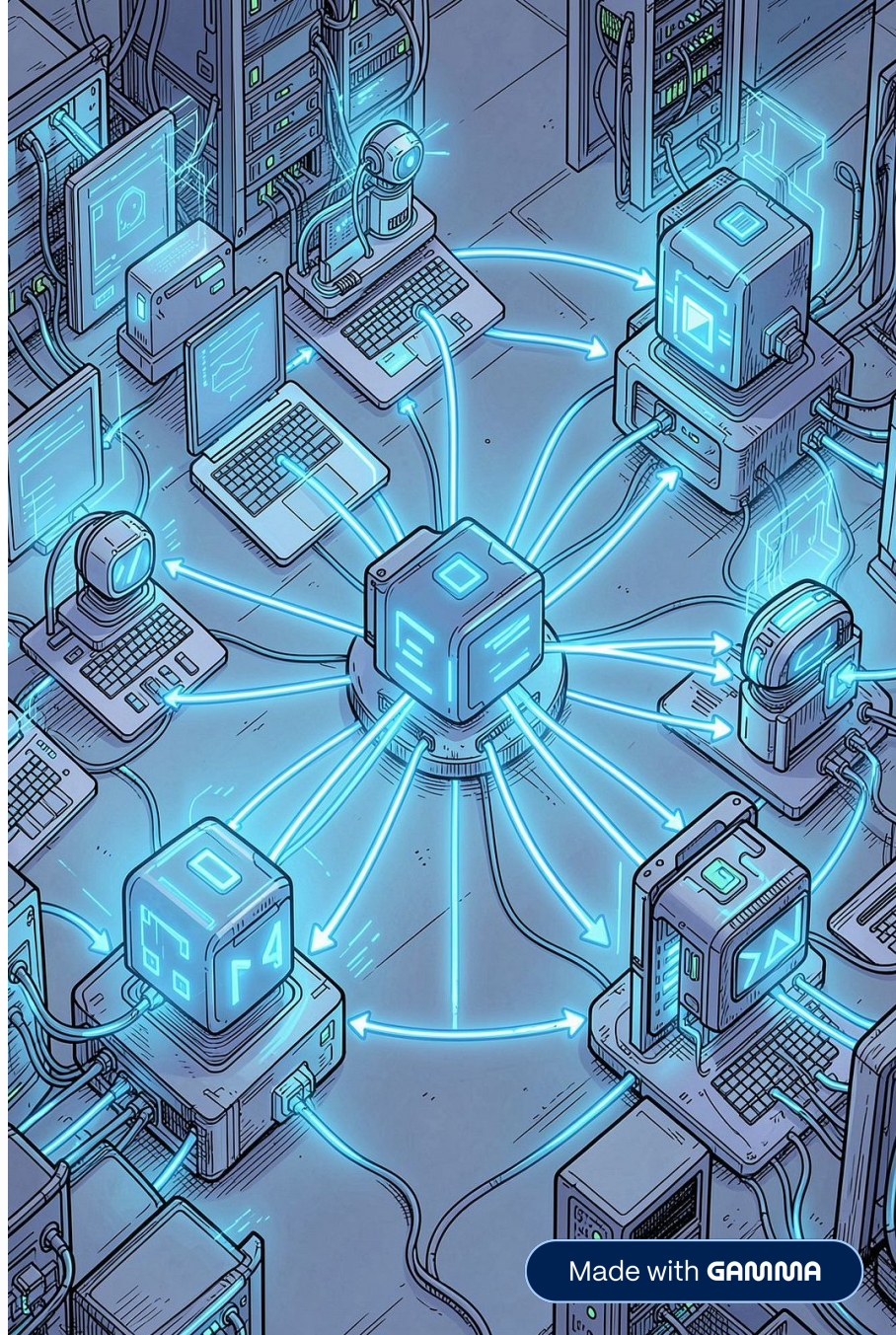
No single point of control

🤝 Collaborative

Agents work together

⚡ Intelligent

Each agent reasons and decides



What is a Multi-Agent System?

A **Multi-Agent System (MAS)** is a system composed of multiple interacting intelligent agents — software entities that perceive their environment, make autonomous decisions, and act to achieve goals.



Agent

An autonomous entity with sensors, reasoning, and actuators

Environment

The shared space in which agents operate and interact

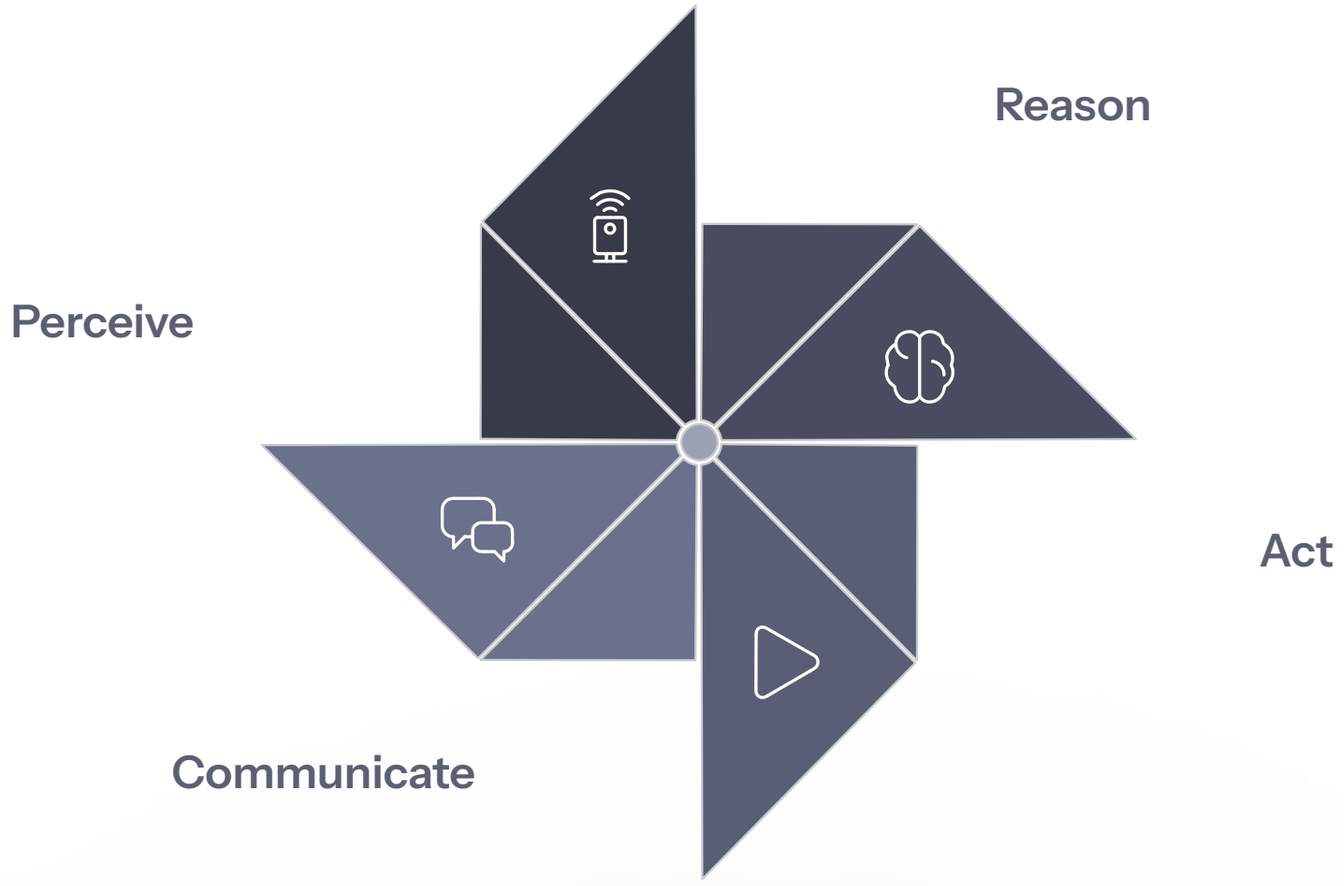
Interaction

Communication, cooperation, or competition between agents

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How Multi-Agent Systems Work

MAS operates through a continuous cycle of perception, reasoning, and action — all happening in parallel across agents.



Agents may **cooperate** (share goals), **compete** (pursue conflicting goals), or **negotiate** to reach agreements — all without centralised control.

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Core Features & Characteristics of MAS

Autonomy

Agents control their own actions without external commands

Reactivity

Agents perceive and respond to environment changes in real time

Proactiveness

Agents take initiative to pursue goals, not just react

Social Ability

Agents communicate and collaborate using shared protocols

Intelligence

Agents use reasoning, learning, or planning to make decisions

Emergence

Complex system-level behaviour arises from simple agent rules

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Important Considerations in MAS Design

1 Coordination & Conflict

How do agents avoid stepping on each other's goals? Mechanisms like negotiation, auctions, and voting are essential.

2 Communication Protocols

Standard languages like **FIPA-ACL** and **KQML** enable structured, meaningful agent-to-agent dialogue.

3 Scalability

The system must perform reliably as the number of agents grows — from tens to thousands.

⚠️ Trust & Security

Malicious agents can disrupt the system; trust models and authentication are critical.

⚖️ Fairness

Resource allocation and task distribution must be equitable among competing agents.

🔄 Adaptability

Agents must handle dynamic, unpredictable environments gracefully.

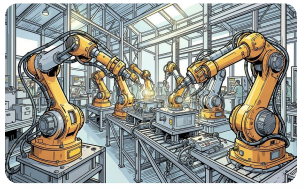
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Common Usages & Applications of MAS



🚗 Smart Transportation

Traffic light coordination, autonomous vehicles, and route optimisation in real time



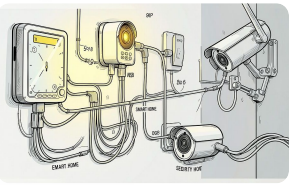
🏭 Industrial Automation

Coordinated robots in manufacturing, warehouse logistics, and supply chain management



📈 Financial Trading

Algorithmic trading bots that negotiate, compete, and adapt in live markets



🏠 Smart Environments

IoT ecosystems where devices autonomously manage energy, security, and comfort

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Conclusion: Key Takeaways on MAS

🔑 Distributed Intelligence

MAS replaces single-point control with collaborative, autonomous agents

🔄 Emergent Behaviour

Complex outcomes arise from simple, local agent interactions

🌍 Real-World Impact

From traffic to trading, MAS powers some of today's most critical systems

Final Thought: As AI and IoT continue to evolve, Multi-Agent Systems will become the backbone of intelligent, adaptive, and scalable solutions across every industry.

Thank you for giving your time

